

sched:AdvanceBookingSchedule

aba:AdvanceBookingActivator

pm:PersistenceManager

fm:FleetMonitor

am:AllocationManager

1 activateDueBookings(now)

2 getDueBookings(now)

3 dueBookings

loop [for each due booking]

4 getCandidates()

5 candidates

alt [reserved robotaxi still eligible]

6 assign(reservedRobotaxi,
request)

7 assignmentConfirmed

[reserved robotaxi no longer eligible]

8 filterAvailable(candidates,
timeWindow)

9 availableCandidates

10 allocate(request,
availableCandidates)

11 selectedRobotaxi

alt [feasible robotaxi found]

reserve(request,
12 selectedRobotaxi,
timeWindow)

13 reservationConfirmed

14 assign(selectedRobotaxi,
request)

15 assignmentConfirmed

[no feasible robotaxi]

16 update(request,
REJECTED)

17 updateConfirmed